



Associative arrays

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0



I was trying to understand the article given at [Storing an array of strings in a command](#) but could not find the (obvious) solution to the following problem:

I have information about composer, title, subtitle and some other metadata indexed by a numerical string id, e.g. 00034, 00035, 02354, 12335 (they are all 5-digit with leading zeros) and I want to store this in an associative array (or perhaps matrix?) and be able to recall that information at various places in a document, e.g.

```
\data{00034}{composer}
```

and also in various 'text manipulations' e.g.

```
\data{00034}{composer}: \data{00034}{title}, \emph{\data{00034}{subtitle}}
```

Is this possible using LaTeX? TIA!

Edit: I forgot to mention that any block of "definitions" or "population" commands are generated by a database export so large blocks of "set" commands as shown in the many fine solutions below is not a problem.

macros

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edited 2 mins ago

asked 10 hours ago

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mf67

668

3

10



Sure thing! How do you input the data? Is it in an external file (like CSV)? Or do you have a manual list somewhere you want to input in your file (like `\addData{<ID>}{<TYPE>}{<VALUE>}` and then use `\addData{01234}{composer}{Someone}`)? – [Werner](#) ♦ 9 hours ago

1



Does this answer your question? [Create an array of variables with string keys](#) – [Marijn](#) 2 hours ago



@Werner I added some information about the 'setting' of data. I export from a database. Your assumptions are correct. – [mf67](#) 1 min ago

3 Answers

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Beethoven: Bagatelle No. 25, *Für Elise*

4



```
\documentclass{article}
```

```
\newcommand\savedata[3]{\expandafter\def\csname data-#1-#2\endcsname{#3}}
```

```
\newcommand\data[2]{\csname data-#1-#2\endcsname}
```

```
\savedata{00034}{composer}{Beethoven}
```

```
\savedata{00034}{title}{Bagatelle No. 25}
```

```
\savedata{00034}{subtitle}{Für Elise}
```

```
\begin{document}
```

```
\data{00034}{composer}: \data{00034}{title}, \emph{\data{00034}{subtitle}}
```

```
\end{document}
```

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answered 9 hours ago



David Carlisle

761k 70 1.6k
2.4k

1 +1 there is simple and straightforward solution using TeX primitives, no strange expl3 language. – [wipet](#) 4 hours ago

@wipet I was thinking of you at the time:-) – [David Carlisle](#) 1 hour ago



This is obviously a job for property lists.

1

See the `\uselists` command to see why: we can map over the known lists; the middle argument is a template where `#1` stands for the current ID.



```
\documentclass{article}
\usepackage{booktabs}

\ExplSyntaxOn

\NewDocumentCommand{\definelist}{mm}
{
  \mflxvii_list_define:nn { #1 } { #2 }
}
\NewDocumentCommand{\addtolist}{mmm}
{
  \mflxvii_list_add:nnn { #1 } { #2 } { #3 }
}
\NewExpandableDocumentCommand{\getitem}{mm}
{
  \mflxvii_list_get:nn { #1 } { #2 }
}
\NewDocumentCommand{\uselists}{mmm}
{
  {% #1 = pre code, #2 = body, #3 = post code
    \mflxvii_list_use:nnn { #1 } { #2 } { #3 }
  }
}

\seq_new:N \g_mflxvii_list_all_seq

% check whether the list already exists
\cs_new_protected:Nn \mflxvii_list_define:nn
{
  \prop_if_exist:cTF { g_mflxvii_list_#1_prop }
  {% already defined
    \msg_error:nnn { mflxvii } { list-exist } { #1 }
  }
  {% define a new one
    % add to the global list of lists
    \seq_gput_right:Nn \g_mflxvii_list_all_seq { #1 }
    % build the list from the given data
    \prop_new:c { g_mflxvii_list_#1_prop }
    \prop_gset_from_keyval:cn { g_mflxvii_list_#1_prop } { #2 }
  }
}

\cs_new_protected:Nn \mflxvii_list_add:nnn
{
  \non if exist:cTF { a mflxvii list #1 non }
```

```

\prop_if_exist:nn { g_mflxvii_list_#1_prop } {
  {% add if existing
    \prop_gput:cnn { g_mflxvii_list_#1_prop } { #2 } { #3 }
  }
  {% list doesn't exist
    \msg_error:nnn { mflxvii } { list-exist } { #1 }
  }
}

\cs_new:Nn \mflxvii_list_get:nn
{
  \prop_item:cn { g_mflxvii_list_#1_prop } { #2 }
}

\cs_new_protected:Nn \mflxvii_list_use:nnn
{
  \group_begin:
  \cs_set:Nn \__mflxvii_list_use_body:n { #2 }
  % pre code
  #1
  % body
  \seq_map_function:NN \g_mflxvii_list_all_seq \__mflxvii_list_use_body:n
  % post code
  #3
  \group_end:
}

\ExplSyntaxOff

\definelist{00034}{
  composer=Beethoven,
  title=Bagatelle No.\@ 25,
  subtitle=Für Elise
}

\definelist{00035}{
  composer=Bob Geldof,
  title=The Great Song of Indifference
}

\addtolist{00035}{subtitle}{I don't mind}

\begin{document}

\getitem{00034}{composer}: \getitem{00034}{title}, \getitem{00034}{subtitle}

\bigskip

\uselists{%
  \begin{tabular}{@{}llll@{}}
  \toprule
  ID & composer & title & subtitle \\

```

```

\midrule
}{ #1 & \getitem{#1}{composer} & \getitem{#1}{title} & \getitem{#1}{subtitle} \
\
}{%
\bottomrule\end{tabular}
}

\end{document}

```

Beethoven: Bagatelle No. 25, Für Elise

ID	composer	title	subtitle
00034	Beethoven	Bagatelle No. 25	Für Elise
00035	Bob Geldof	The Great Song of Indifference	I don't mind

Seems you have an array of records so that each array-element itself is a record which has data-fields.

1

You can define a control sequence for each data-item:

```
\documentclass{article}

\ExplSyntaxOn
\cs_new:Npn \StoreData #1#2#3 {
  \cs_new:cpn {ArrayItem:{#1}_RecordField:{#2}} {#3}
}
\cs_new:Npn \Data #1#2 {
  \cs_if_exist_use:cF {ArrayItem:{#1}_RecordField:{#2}} {--
data~not~available--}
}
\ExplSyntaxOff

%-----
\StoreData {00034}{composer}{Beethoven}
\StoreData {00034}{title}{Bagatelle No. 25}
\StoreData {00034}{subtitle}{Für Elise}
%-----
\StoreData {00035}{composer}{Bob Geldof}
\StoreData {00035}{title}{The Great Song of Indifference}
\StoreData {00035}{subtitle}{I don't mind}
%-----

\begin{document}

\Data{00034}{composer}\par
\Data{00034}{title}\par
\Data{00034}{subtitle}\par
\Data{00035}{composer}\par
\Data{00035}{title}\par
\Data{00035}{subtitle}\par
\Data{00035}{Unknown}\par
\Data{00036}{subtitle}

\end{document}
```

You can use expl3's package l3prop and maintain a property list where names of properties are combined from names of array-elements and names of array-elements' fields:

```
\documentclass{article}

\ExplSyntaxOn
```

```

\prop_new:N \g__MyModule_Array_prop
\cs_new:Npn \AddArrayProperty #1#2#3 {
  \prop_gput:Nnn \g__MyModule_Array_prop {ArrayItem:{#1}_RecordField:{#2}} {#3}
}
\cs_new:Npn \Data #1#2 {
  \prop_if_in:NnTF
    \g__MyModule_Array_prop
    {ArrayItem:{#1}_RecordField:{#2}}
    { \prop_item:Nn \g__MyModule_Array_prop {ArrayItem:{#1}_RecordField:{#2}} }
    {--data~not~available--}
}
\ExplSyntaxOff

%-----
\AddArrayProperty {00034}{composer}{Beethoven}
\AddArrayProperty {00034}{title}{Bagatelle No. 25}
\AddArrayProperty {00034}{subtitle}{Für Elise}
%-----
\AddArrayProperty {00035}{composer}{Bob Geldof}
\AddArrayProperty {00035}{title}{The Great Song of Indifference}
\AddArrayProperty {00035}{subtitle}{I don't mind}
%-----

\begin{document}

\Data{00034}{composer}\par
\Data{00034}{title}\par
\Data{00034}{subtitle}\par
\Data{00035}{composer}\par
\Data{00035}{title}\par
\Data{00035}{subtitle}\par
\Data{00035}{Unknown}\par
\Data{00036}{subtitle}

\end{document}

```

You can try nesting `\str_case:nnF` :

```

\documentclass{article}

\newcommand\MyArrayOfRecords{%
  {00034}{%
    {composer}{Beethoven}%
    {title}{Bagatelle No. 25}%
    {subtitle}{Für Elise}%
  }%
  {00035}{%
    {composer}{Bob Geldof}%
    {title}{The Great Song of Indifference}%
  }%
}

```

```

        {subtitle}{I don't mind}%
    }%
}%

\ExplSyntaxOn
\cs_new:Nn \__MyStuff_Check_Item:nnn {
    \quark_if_no_value:nTF
        {#1}
        {--#2~not~available--}
        { \str_case:nnF {#3}{#1}{--#3~not~available~in~#2--} }
}
\cs_new:Npn \Data #1#2 {
    \exp_args:Nf
        \__MyStuff_Check_Item:nnn
        { \exp_args:Nno \str_case:nnF {#1} { \MyArrayOfRecords } {\exp_not:N
\q_no_value} }
        {#1}
        {#2}
}
\ExplSyntaxOff

\begin{document}

\Data{00034}{composer}\par
\Data{00034}{title}\par
\Data{00034}{subtitle}\par
\Data{00035}{composer}\par
\Data{00035}{title}\par
\Data{00035}{subtitle}\par
\Data{00035}{Unknown}\par
\Data{00036}{subtitle}

\end{document}

```

In case the data is available as .csv-list, the package [datatool](#) might be of interest.